

Geo-Spatial Analysis

Week- 3

Task-3 (DS-3)

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1. Introduction:

This project performs a geo-spatial analysis by combining NHS Talking Therapies (IAPT) data with the Index of Multiple Deprivation (IMD) dataset across England.

The analysis aims to identify:

- highly deprived regions.
- areas with lower mental health service accessibility.
- high-need low-service areas.

Spatial joins and geographic aggregation techniques were used to connect deprivation indicators with NHS regional service data at the Integrated Care Board (ICB) level. Choropleth maps were created to visualize deprivation patterns, service access rates, and underserved regions across England.

2. Dataset:

- **NHS Talking Therapies (IAPT) Dataset:**

Description

This dataset contains information related to NHS Talking Therapies mental health services across England. It includes service activity, treatment access, recovery outcomes, and waiting time indicators reported at different NHS geographic levels.

Key Features Used

- Referrals received
- Accessing services
- Recovery rate
- Mean waiting time
- Waiting over 90 days
- SubICB geographic codes

The dataset was used to analyze:

- mental health service accessibility.
- treatment performance
- regional service inequalities.

	REPORTING_PERIOD_START	REPORTING_PERIOD_END	GROUP_TYPE	ORG_CODE1	ORG_NAME1	ORG_CODE2	ORG_NAME2	MEASURE_ID	MEASURE_NAME	MEASURE_VALUE_SUPPRESSED
0	2026-02-01	2026-02-28	England	all	all SubICBs	all	all Providers	M001	Count_ReferralsReceived	151810
1	2026-02-01	2026-02-28	England	all	all SubICBs	all	all Providers	M002	Count_SelfReferrals	103623
2	2026-02-01	2026-02-28	England	all	all SubICBs	all	all Providers	M003	Count_GPReferrals	13041
3	2026-02-01	2026-02-28	England	all	all SubICBs	all	all Providers	M004	Count_HealthVisitorReferrals	445
4	2026-02-01	2026-02-28	England	all	all SubICBs	all	all Providers	M005	Count_OtherPrimaryCareReferrals	5750

- **IMD Dataset:**

Description

The IMD dataset measures relative deprivation across small geographic regions in England called Lower Layer Super Output Areas (LSOAs). It combines multiple socioeconomic indicators such as:

- income
- employment
- education
- crime
- housing
- health deprivation

Key Features Used

- LSOA code
- IMD rank
- IMD decile
- Standardized IMD score

The dataset was used to identify:

- Highly deprived regions.
- Areas with potentially higher mental health support needs.

	LSOA code (2021)	LSOA name (2021)	Local Authority District code (2024)	Local Authority District name (2024)	Index of Multiple Deprivation (IMD) Rank (where 1 is most deprived)	Index of Multiple Deprivation (IMD) Decile (where 1 is most deprived 10% of LSOAs)
0	E01000001	City of London 001A	E09000001	City of London	26525	8
1	E01000002	City of London 001B	E09000001	City of London	31203	10
2	E01000003	City of London 001C	E09000001	City of London	25913	8
3	E01000005	City of London 001E	E09000001	City of London	14807	5
4	E01000006	Barking and Dagenham 016A	E09000002	Barking and Dagenham	10917	4

- **LSOA to ICB Look up Dataset:**

Description

This lookup dataset connects small geographic areas (LSOAs) with NHS administrative regions including:

- SubICBs
- Integrated Care Boards (ICBs)

Key Features Used

- LSOA21CD
- SICBL23CDH
- ICB23CD
- ICB23NM

The lookup table was used to:

- Spatially join IMD data with NHS regions
- Map LSOAs to ICB geography.
- Support geographic aggregation and merging.

	LSOA21CD	LSOA21NM	SICBL23CD	SICBL23CDH	SICBL23NM	ICB23CD	ICB23CDH	ICB23NM	LAD23CD	LAD23NM	ObjectId
0	E01011954	Hartlepool 001A	E38000247	16C	NHS North East and North Cumbria ICB - 16C	E54000050	QHM	NHS North East and North Cumbria Integrated Ca...	E06000001	Hartlepool	1
1	E01011969	Hartlepool 001B	E38000247	16C	NHS North East and North Cumbria ICB - 16C	E54000050	QHM	NHS North East and North Cumbria Integrated Ca...	E06000001	Hartlepool	2
2	E01011970	Hartlepool 001C	E38000247	16C	NHS North East and North Cumbria ICB - 16C	E54000050	QHM	NHS North East and North Cumbria Integrated Ca...	E06000001	Hartlepool	3
3	E01011971	Hartlepool 001D	E38000247	16C	NHS North East and North Cumbria ICB - 16C	E54000050	QHM	NHS North East and North Cumbria Integrated Ca...	E06000001	Hartlepool	4
4	E01033465	Hartlepool 001F	E38000247	16C	NHS North East and North Cumbria ICB - 16C	E54000050	QHM	NHS North East and North Cumbria Integrated Ca...	E06000001	Hartlepool	5

- **Integrated Care Board Boundary Shapefile:**

Description

This shapefile contains geographic boundary polygons for Integrated Care Boards across England.

The shapefile was used to:

- Create choropleth maps.
- Visualise deprivation patterns.
- Display service accessibility across England.
- Identify high-need low-service regions spatially.

	ICB23CD	ICB23NM
0	E54000008	NHS Cheshire and Merseyside Integrated Care Board
1	E54000010	NHS Staffordshire and Stoke-on-Trent Integrate...
2	E54000011	NHS Shropshire, Telford and Wrekin Integrated ...
3	E54000013	NHS Lincolnshire Integrated Care Board
4	E54000015	NHS Leicester, Leicestershire and Rutland Inte...

3. Preparing NHS data for Spatial Analysis:

- Converted service values into numeric format.
- Mapped SubICB codes to ICB names using the geographic lookup table.
- Extracted key mental health service indicators.
- Aggregated service metrics to ICB level.
- Calculated an Access rate indicator for comparing service provision across regions.

Rows matched to ICB: 23,002 / 24,521

Service table: 42 ICBs

	ICB23NM	Referrals	Accessing_Services	Recovery_Rate	Mean_Wait_Days	Waiting_Over90	Access_Rate
0	NHS Bath and North East Somerset, Swindon and ...	1965.0	1350.0	46.0	11.7	20.0	68.70
1	NHS Bedfordshire, Luton and Milton Keynes Inte...	2640.0	1390.0	47.0	35.3	265.0	52.65
2	NHS Birmingham and Solihull Integrated Care Board	4340.0	2645.0	48.0	19.7	400.0	60.94
3	NHS Black Country Integrated Care Board	3025.0	1740.0	44.0	49.2	1115.0	57.52
4	NHS Bristol, North Somerset and South Gloucest...	2255.0	1205.0	50.0	32.2	20.0	53.44

4. Preparing and Standardizing IMD Deprivation data:

- Renamed long column titles into shorter and cleaner variable.
- Converted IMD ranks into an intuitive deprivation score.
- Retained only the variables required for spatial analysis.
- Verified the range and structure of the processed deprivation dataset.

```
IMD LSOAs: 33,755
IMD Score range: 0.00 (least deprived) - 100.00 (most deprived)
```

	LSOA21CD	IMD_Score	IMD_Rank	IMD_Decile
0	E01000001	21.4220	26525	8
1	E01000002	7.5633	31203	10
2	E01000003	23.2351	25913	8
3	E01000005	56.1369	14807	5
4	E01000006	67.6611	10917	4

5. Spatial Joining:

- Matched LSOA codes between the IMD dataset and lookup table.
- Added SubICB and ICB geographic indentifiers.
- Verified geographic coverage across Integrated Care Boards (ICB).
- Created a combined dataset for later aggregation and spatial analysis.

```
IMD + Lookup rows: 33,755
ICBs covered: 42
```

	LSOA21CD	IMD_Score	IMD_Rank	IMD_Decile	SICBL23CDH	ICB23CD	ICB23NM
0	E01000001	21.4220	26525	8	A3A8R	E54000029	NHS North East London Integrated Care Board
1	E01000002	7.5633	31203	10	A3A8R	E54000029	NHS North East London Integrated Care Board
2	E01000003	23.2351	25913	8	A3A8R	E54000029	NHS North East London Integrated Care Board
3	E01000005	56.1369	14807	5	A3A8R	E54000029	NHS North East London Integrated Care Board
4	E01000006	67.6611	10917	4	A3A8R	E54000029	NHS North East London Integrated Care Board

6. Aggregating Deprivation Data to ICB level:

- Mean IMD score.
- Median IMD score.
- Mean IMD decile.
- Number of LSOAs within each ICB.

ICB-level IMD table: 42 ICBs

	ICB23NM	IMD_Score_Mean	IMD_Score_Median	IMD_Decile_Mean	n_LSOAs
0	NHS Bath and North East Somerset, Swindon and ...	36.1567	31.6723	6.8524	569
1	NHS Bedfordshire, Luton and Milton Keynes Inte...	46.4414	42.8885	5.8552	587
2	NHS Birmingham and Solihull Integrated Care Board	67.4908	78.5247	3.7680	793
3	NHS Black Country Integrated Care Board	64.4896	75.4111	4.0443	723
4	NHS Bristol, North Somerset and South Gloucest...	40.8421	36.5812	6.3838	581

7. Combining Mental Health Service Data with Deprivation Indicators:

- Mental health service provision metrics.
- Deprivation indicators.
- Regional NHS geography.

	ICB23NM	Referrals	Access_Rate	Recovery_Rate	IMD_Score_Mean
0	NHS Bath and North East Somerset, Swindon and ...	1965.0	68.70	46.0	36.1567
1	NHS Bedfordshire, Luton and Milton Keynes Inte...	2640.0	52.65	47.0	46.4414
2	NHS Birmingham and Solihull Integrated Care Board	4340.0	60.94	48.0	67.4908
3	NHS Black Country Integrated Care Board	3025.0	57.52	44.0	64.4896
4	NHS Bristol, North Somerset and South Gloucest...	2255.0	53.44	50.0	40.8421

8. Identifying High Need and Low Service Areas:

Classification Logic:

- **High Need** if they belonged to the top 25% most deprived ICBs.
- **Low Service** if they fell within the bottom 25% for key service indicators such as referrals, access rates, or recovery rates.

Underserved ICBs identified: 7 out of 42

HIGH-NEED / LOW-SERVICE ICBs:

	ICB23NM	IMD_Score_Mean	Referrals	Access_Rate	Recovery_Rate	Mean_Wait_Days	Unmet_Need_Score
NHS Greater Manchester Integrated Care Board		60.1235	12210.0	49.51	42.90	38.98	74.35
NHS Black Country Integrated Care Board		64.4896	3025.0	57.52	44.00	49.20	73.44
NHS North East London Integrated Care Board		62.8046	5440.0	55.33	49.00	32.50	62.70
NHS North Central London Integrated Care Board		60.1696	4120.0	56.43	46.00	25.90	62.50
NHS North West London Integrated Care Board		57.6900	6790.0	54.20	45.00	17.60	61.44
NHS Lancashire and South Cumbria Integrated Care Board		55.5904	4800.0	59.69	43.25	25.75	60.33
NHS Cornwall and the Isles of Scilly Integrated Care Board		57.8143	1280.0	56.64	53.00	25.30	50.39

9. Preparing Spatial Boundary Data for Mapping:

- Geographic boundaries.
- Deprivation indicators.
- Mental health service metrics.
- Underserved area classifications.

Spatial dataframe: 42 ICBs

Underserved ICBs in map: 7

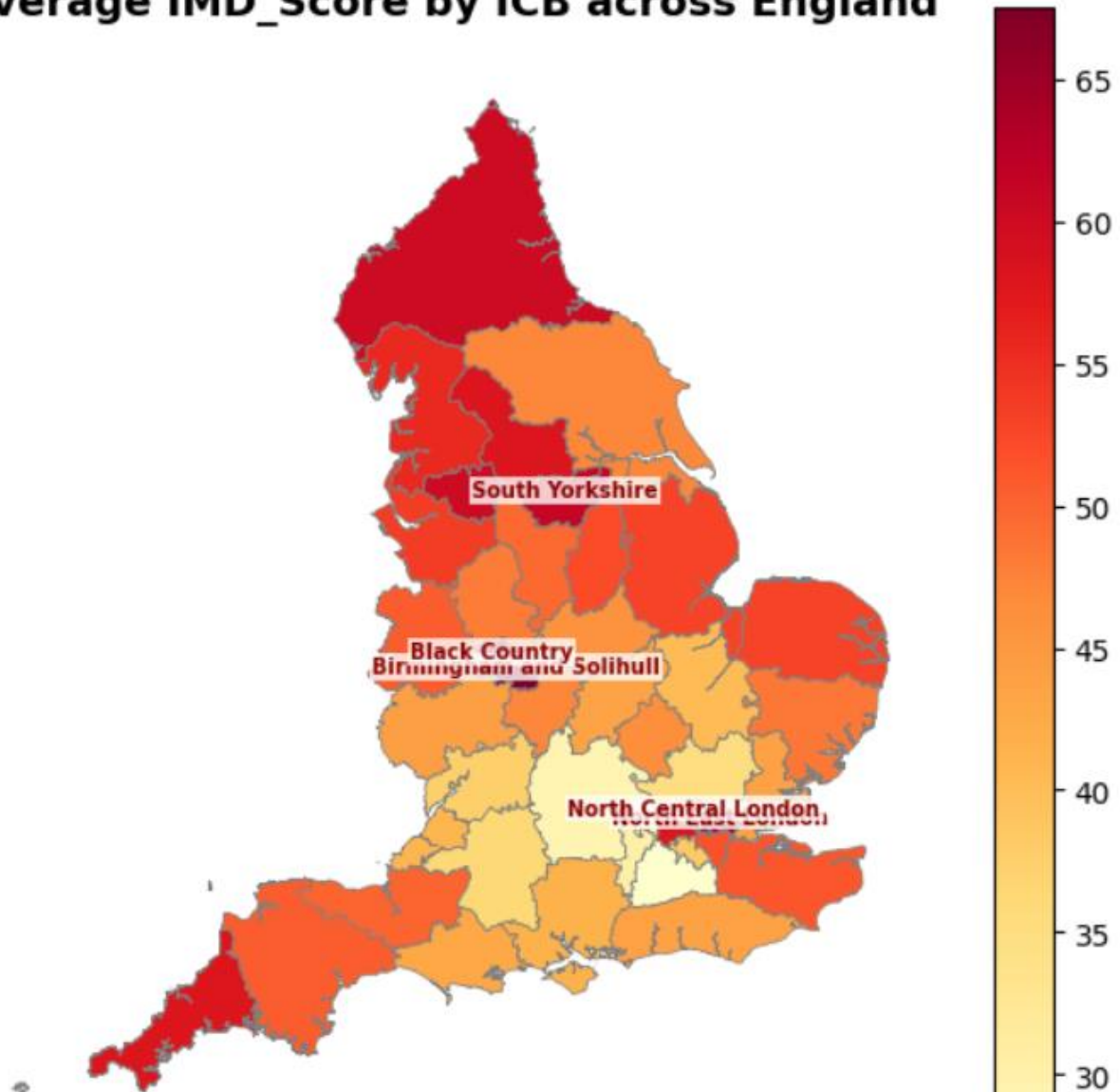
	ICB23NM	IMD_Score_Mean	Access_Rate	underserved
0	NHS Cheshire and Merseyside Integrated Care Board	53.5083	62.79	False
1	NHS Staffordshire and Stoke-on-Trent Integrate...	48.0824	60.50	False
2	NHS Shropshire, Telford and Wrekin Integrated ...	50.7043	40.74	False
3	NHS Lincolnshire Integrated Care Board	52.9343	57.99	False
4	NHS Leicester, Leicestershire and Rutland Inte...	45.7176	65.54	False

10. Visualizations:

10.1 Average IMD score across England

- Darker regions represent higher deprivation.
- Lighter regions represent lower deprivation.

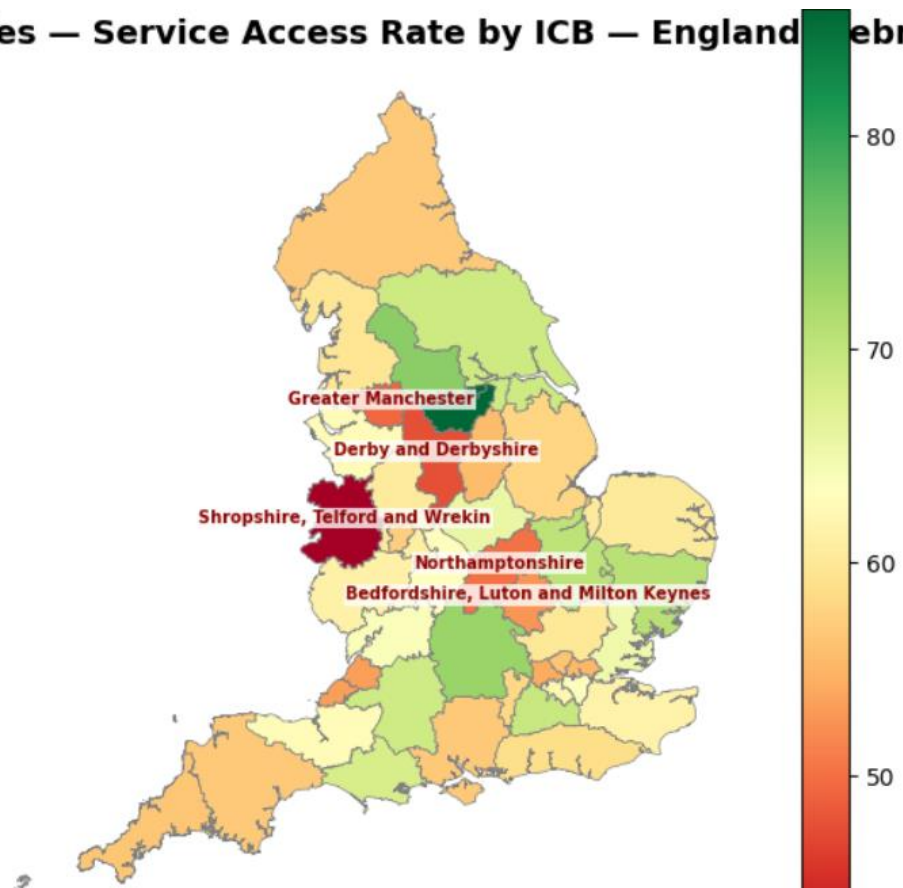
Average IMD_Score by ICB across England



10.2 NHS talking therapies Service Access across England

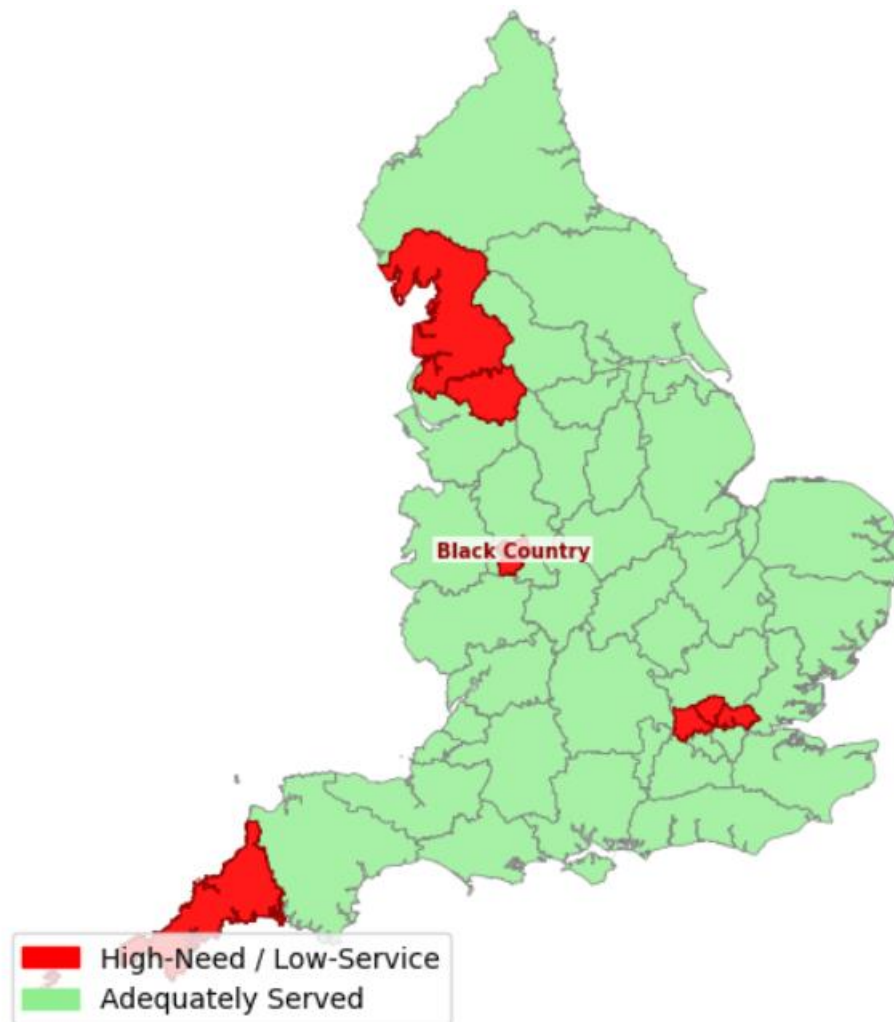
- Green regions represent higher service access.
- Red regions represent lower service access.

NHS Talking Therapies — Service Access Rate by ICB — England (February 2026)



10.3 Identifying High Need Low Service Areas

- High Need / Low Service if they belonged to highly deprived regions while also showing weaker service indicators.
- Adequately Served if service performed was comparatively better.



11. Key Findings:

- The geo-spatial analysis identified multiple Integrated Care Boards (ICBs) with high deprivation levels and comparatively weaker NHS Talking Therapies service performance.
- Regions with higher IMD scores generally showed:
 - lower access rates,
 - longer waiting times,
 - and comparatively lower recovery outcomes.
- NHS Greater Manchester Integrated Care Board recorded the highest unmet need score, indicating significant pressure on mental health service accessibility and outcomes.
- Several London-based ICBs, including North East London and North Central London, demonstrated high deprivation alongside below-average service indicators.
- The average access rate among priority high-need ICBs (55.6%) was lower than the England national average (60.9%).
- Average recovery performance in high-priority regions (46.2%) was also slightly below the national average recovery rate (47.5%).
- Waiting times in several high-need regions exceeded the England average of 26.5 days, suggesting increased service demand and potential capacity limitations.
- The analysis highlighted clear spatial inequalities in mental healthcare provision across England.
- Combining NHS Talking Therapies data with IMD deprivation indices proved effective in identifying underserved regions and potential areas requiring targeted mental health interventions.

High-Need / Low-Service ICBs

	ICB Name	IMD Score	Access Rate (%)	Recovery Rate (%)	Average Wait (Days)	Unmet Need Score
	NHS Greater Manchester Integrated Care Board	60.1	49.5	42.9	39.0	74.4
	NHS Black Country Integrated Care Board	64.5	57.5	44.0	49.2	73.4
	NHS North East London Integrated Care Board	62.8	55.3	49.0	32.5	62.7
	NHS North Central London Integrated Care Board	60.2	56.4	46.0	25.9	62.5
	NHS North West London Integrated Care Board	57.7	54.2	45.0	17.6	61.4
	NHS Lancashire and South Cumbria Integrated Care Board	55.6	59.7	43.2	25.8	60.3
	NHS Cornwall and the Isles of Scilly Integrated Care Board	57.8	56.6	53.0	25.3	50.4

National Comparisons

Average England access rate : 60.9%
 Average England recovery rate : 47.5%
 Average England waiting time : 26.5 days

Priority ICB average access rate : 55.6%
 Priority ICB average recovery rate : 46.2%